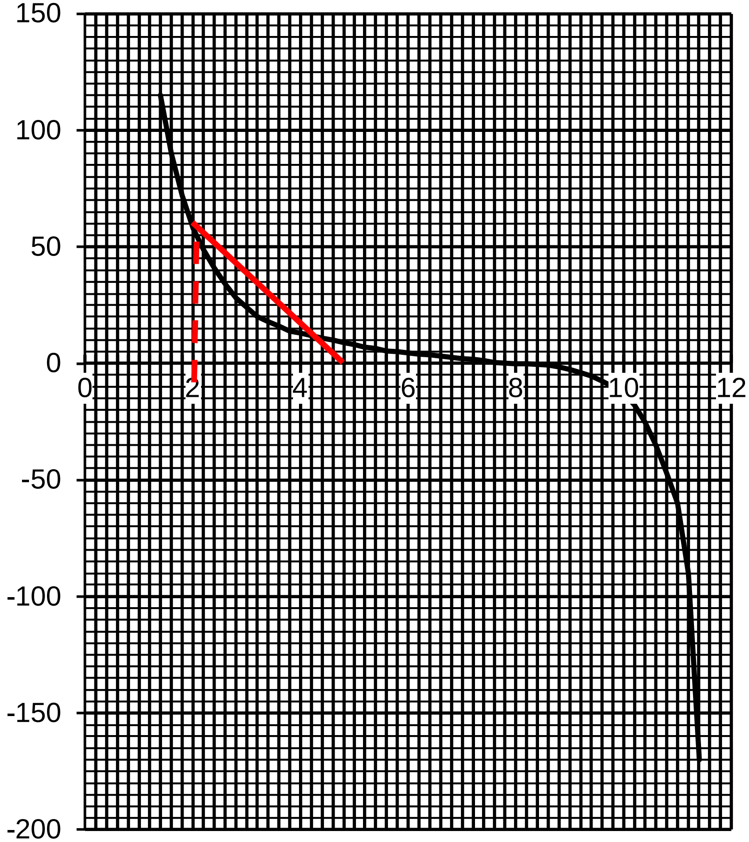
	where k is the constant of proportionality.	
5(a)	Zero electric field strengths in sphere A (between x = 0 and x = 1.4 cm) and in sphere B (between x = 11.4 and x = 12.0 cm)	[1]
(b)(i)	The net field strength is zero at a point between the spheres and this shows that electric fields are in opposite directions in the region between the two spheres. This shows that the polarity of the two charges are the same. Hence, since sphere A is positively charged, sphere B must also be positively charged.	[1] [1] conclude that both charges are the same polarity
(ii)	At x = 0.08 m, the electric field strength due to sphere A cancels out the electric field strength due to sphere B. $E_A = E_B$ $$\frac{Q_A}{4\pi\varepsilon_o(0.08)^2}=\frac{Q_B}{4\pi\varepsilon_o(0.04)^2}$$ $$\frac{Q_A}{Q_B}=\left(\frac{0.08}{0.04}\right)^2=4$$	[1] correct expression of field strength [1] for correct answer
(c)(i)	The electric field strength is negative of the electric potential gradient, i.e. $E=-\frac{dV}{dx}$	[1]

(ii)	 $V = -\int_{x=2cm}^{x=8cm} E\, dx$ Hence, change in potential from *x* = 2.0 cm to *x* = 8.0 cm, ΔV = area under *E*−*x* graph (from *x* = 2.0 cm to *x* = 8.0 cm) $\Delta V = \frac{1}{2}\left(2.6\times10^{-2}\right)\left(60\times10^{6}\right) = 7.8\times10^{5}\ \text{V}$ Work done $= \left(7.8\times10^{5}\right)\left(2.0\times10^{-6}\right) = 1.6\ \text{J}$	[1] change in potential [1] for correct working for work done, range of 1.40 J to 1.76 J
6(c)(i)	Total resistance	